

PHYSICS (042)

Reconstructed Question Paper — 2025

Reconstructed from the submitted answer booklet. Numerical data visible in the working has been retained where legible; exact MCQ stems and diagrams are not invented.

Section A — Multiple Choice / Assertion–Reason

Q.1–Q.16 Answer the following objective questions. Identifiable topics include:

1. electromagnetic radiation,
2. photoelectric effect,
3. nuclear/atomic physics,
4. wave optics,
5. de Broglie wavelength,
6. X-rays,
7. assertion–reason questions from modern physics and optics.

Section B — Numerical and Short Answer Questions

Q.17 Light of frequency approximately

$$\nu = 6.8 \times 10^{14} \text{ Hz}$$

is incident on a metal whose threshold frequency is approximately

$$\nu_0 = 3.6 \times 10^{14} \text{ Hz}.$$

Using Einstein's photoelectric equation

$$h\nu = h\nu_0 + K_{\max},$$

calculate the stopping potential.

Q.18 Two coherent sources produce interference. If the path difference corresponds to a phase difference of

$$\phi = \frac{2\pi}{3},$$

determine the intensity at the point in terms of the intensity I_0 of either source.

Q.19 Explain the relation between the phase difference and path difference in interference:

$$\phi = \frac{2\pi}{\lambda} \Delta x.$$

Q.20 A nuclear reaction involving uranium produces xenon, strontium and neutrons. Calculate the mass defect and binding energy using

$$E = \Delta mc^2.$$

Q.21 The resistance of a wire is $10\ \Omega$ at 25°C and $10.5\ \Omega$ at 125°C . Assuming a linear temperature dependence,

$$R_T = R_0(1 + \alpha\Delta T),$$

calculate the temperature coefficient of resistance and the resistance at 425°C .

Section C — Semiconductor Electronics and Electrostatics

Q.22 Explain the distinction between conductors, insulators and semiconductors using energy-band diagrams. Explain the behaviour of an intrinsic semiconductor at:

- (a) $T = 0\ \text{K}$,
- (a) room temperature.

Explain electron–hole pair generation.

Q.23 Explain the capacitance of a parallel-plate capacitor:

$$C = \frac{\varepsilon_0 A}{d}.$$

Derive the equivalent capacitance for:

- (a) two capacitors connected in series,
- (a) two capacitors connected in parallel.

Q.24 In Young’s double-slit experiment, light of wavelengths $500\ \text{nm}$ and $600\ \text{nm}$ is used with slit separation $d = 1\ \text{mm}$. Determine:

- (a) the position of the first bright fringe for each wavelength,
- (a) the distance between the first bright fringes,
- (a) the condition for coincident bright fringes,
- (a) the first coincidence order,
- (a) the minimum distance from the central maximum at which coincidence occurs.

Section D — Semiconductor Rectifiers

Q.25 Distinguish between a half-wave rectifier and a full-wave rectifier with respect to:

- (i) number of half cycles rectified,
- (i) output voltage,
- (i) number of diodes used,
- (i) efficiency/ripple where relevant.

Q.26 Explain the working of a full-wave rectifier and show how both half cycles of the input AC voltage contribute to the output.

The original circuit diagrams, exact numerical tables and some later questions are not sufficiently legible in the submitted answer booklet.